

1.

Create a **Departments** table with:

- **department_id** as the Primary Key
- **department_name**

```
create table Department (department_id int primary key,  
department_name varchar(20));
```

Then create an **Employees** table where:

- **employee_id** is the Primary Key
- **department_id** references the **Departments** table.

```
create table Employees (employee_id int, foreign key (employee_id)  
references Departments (department_id));
```

2.

A college wants to store student information and course details.

- Create a **Students** table with a Primary Key.
- Create an **Enrollments** table that uses a Foreign Key to refer to students.

```
create table students (std_id int primary key, std_name varchar(20),  
std_marks int);
```

```
create table Enrollments (enroll_id int, foreign key (enroll_id) references  
students (std_id));
```

3.

Design two tables:

- **Authors**
- **Books**

Each book should belong to one author. Write the SQL statements.

```
create table Authors (author_id int primary key);
```

```
create table Books (book_id int, foreign key (book_id) references Authors  
(author_id));
```

4.

Create a **Customers** table and an **Orders** table such that each order belongs to a customer using a Foreign Key.

```
create table Customers (customer_id int primary key);
```

```
create table Orders (ord_id int, foreign key (ord_id) references Customers  
(customer_id));
```

5.

A hospital stores doctor information separately from appointments.

Write SQL statements to:

- Create a **Doctors** table.
- Create an **Appointments** table where each appointment references a doctor.

```
create table Doctors (doctor_id int primary key, doctor_name varchar(20)  
not null, qualification varchar(20) not null, phone BIGINT not null);
```

```
create table Appointments (appoint_id int, appoint_reason varchar(50),  
foreign key (appoint_id) references Doctors (doctor_id));
```

6.

Write SQL statements to create:

- Categories
- Products

Each product must belong to exactly one category using a Foreign Key.

```
create table Categories (cate_id int primary key, cat_name varchar(20) not null);
```

```
create table Products (prod_id int, prod_name varchar(50) not null, foreign key  
(prod_id) references Categories (cate_id));
```

7.

A company has multiple projects.

Create:

- Projects
- Employees

Assume each employee is assigned to one project.

```
create table Projects (project_id int primary key, project_name varchar(20),  
project_guide varchar(20));
```

```
create table Employees (employ_id int, employ_name varchar(20),  
employ_project varchar(20), foreign key (employ_id) references Projects  
(project_id));
```

8.

Design a database for an online library:

- Members
- BorrowedBooks

Each borrowed book record should refer to a member using a Foreign Key.

```
create table Members (mem_id int primary key, mem_name varchar(20), phone  
bigint not null, place varchar(50));
```

Create table BorrowedBooks (book_id int, book_name varchar(20), book_cost int, foreign key (book_id) references Members (mem_id));

9.

Create a **Schools** table and a **Teachers** table.

Each teacher should be associated with one school.

create table Schools

create table Schools (scl_id int primary key, scl_name varchar(50), phone BIGINT not null, location varchar(20));

create table Teachers (tech_id int, tech_name varchar(20), dept varchar(20), phone BIGINT not null, foreign key (tech_id) references Schools (scl_id));

10.

Write SQL statements to create:

- **Suppliers**
- **Items**

Each item is supplied by one supplier.

create table Suppliers (suplr_id int auto_increment primary key, suplr_name varchar(20), phone BIGINT not null, location varchar(20));

create table Items (itm_id int, itm_name varchar(20), foreign key (itm_id) references Suppliers (suplr_id));

11.

Create a **Countries** table and a **Cities** table.

Each city should reference its country using a Foreign Key.

```
create table Countries (country_code int primary key, country_name  
varchar(30), capital varchar(20), population BIGINT );
```

```
create table Cities (city_code int, city_name varchar(20), foreign key  
(city_code) references Country (country_code));
```

12.

A movie booking system stores theatre information separately.

Create:

- Theatres
- Shows

Each show belongs to one theatre.

```
create table Theatres (theatre_code int primary key, theatre_name varchar(30),  
location varchar(30) not null, contact BIGINT);
```

```
create table Shows (show_code int , show_name varchar(20), location  
varchar(20), foreign key (show_code) references Theatres (theatre_code));
```

13.

Create a Publishers table and a Magazines table.

Each magazine should reference its publisher.

```
create table Publishers (pub_id int primary key, pub_name varchar(30), location  
varchar(30));
```

```
create table Magazines (magaz_id int, magaz_name varchar(30), magaz_cost  
int, foreign key (magaz_id) references Publishers (pub_id));
```

14.

A university stores department details and faculty details.

Write SQL statements where every faculty member belongs to one department.

```
create table department (dept_id int primary key, dept_name varchar(30));
```

```
create table faculty (fac_id int, fac_name varchar(20), salary int, foreign key  
(fac_id) references department (dept_id));
```

15.

Create a **Brands** table and a **Mobiles** table.

Each mobile should be linked to a brand using a Foreign Key.

```
create table Brands (brand_code int primary key, brand_name varchar(20));
```

```
create table Mobiles (mob_id int, mob_name varchar(30), made_by varchar(30),  
foreign key (mob_id) references Brands (brand_id));
```

16.

A food delivery application has:

- **Restaurants**
- **MenuItems**

Each menu item should belong to a restaurant.

```
create table Restaurants (restaurant_code int primary key, restaurant_name  
varchar(30), location varchar(30));
```

```
create table MenuItems (menu_id int, menu_name varchar(30), foreign key  
(menu_id) references Restaurants (restaurant_code));
```

17.

Create the following tables:

- **Courses**
- **Assignments**

Each assignment belongs to one course.

```
create table Courses (course_id int primary key, course_name varchar(20));
```

```
create table Assignments (assign_id int, assign_name varchar(20), foreign key  
(assign_id) references Courses (course_id));
```

18.

A transport company stores:

- Drivers
- Trips

Each trip should reference one driver.

```
create table Drivers (driv_id int primary key, driv_name varchar(20));
```

```
create table Trips (trip_code int, trip_name varchar(30), location varchar(30),  
foreign key (trip_code) references Drivers (driv_id));
```

19.

Write SQL statements to create:

- Parents
- Children

Each child must reference a parent using a Foreign Key.

```
create table Parents (parent_id int primary key, parent_name varchar(20));
```

```
create table Children (child_id int, child_name varchar(20), foreign key  
(child_id) references Parents (parent_id));
```

20.

A shopping application contains:

- Warehouses
- Products

Each product is stored in one warehouse.

Create both tables with appropriate Primary Key and Foreign Key constraints.

```
create table Warehouses (warehouse_id int primary key, warehouse_name  
varchar(30));
```

```
create table Products (prod_id int, prod_name varchar(30),foreign key (prod_id)  
references (warehouse_id));
```

These questions are suitable for practicing: